School of Computer Science and Engineering

CSE1004 Networks and Communication

Lab Experiment – 1

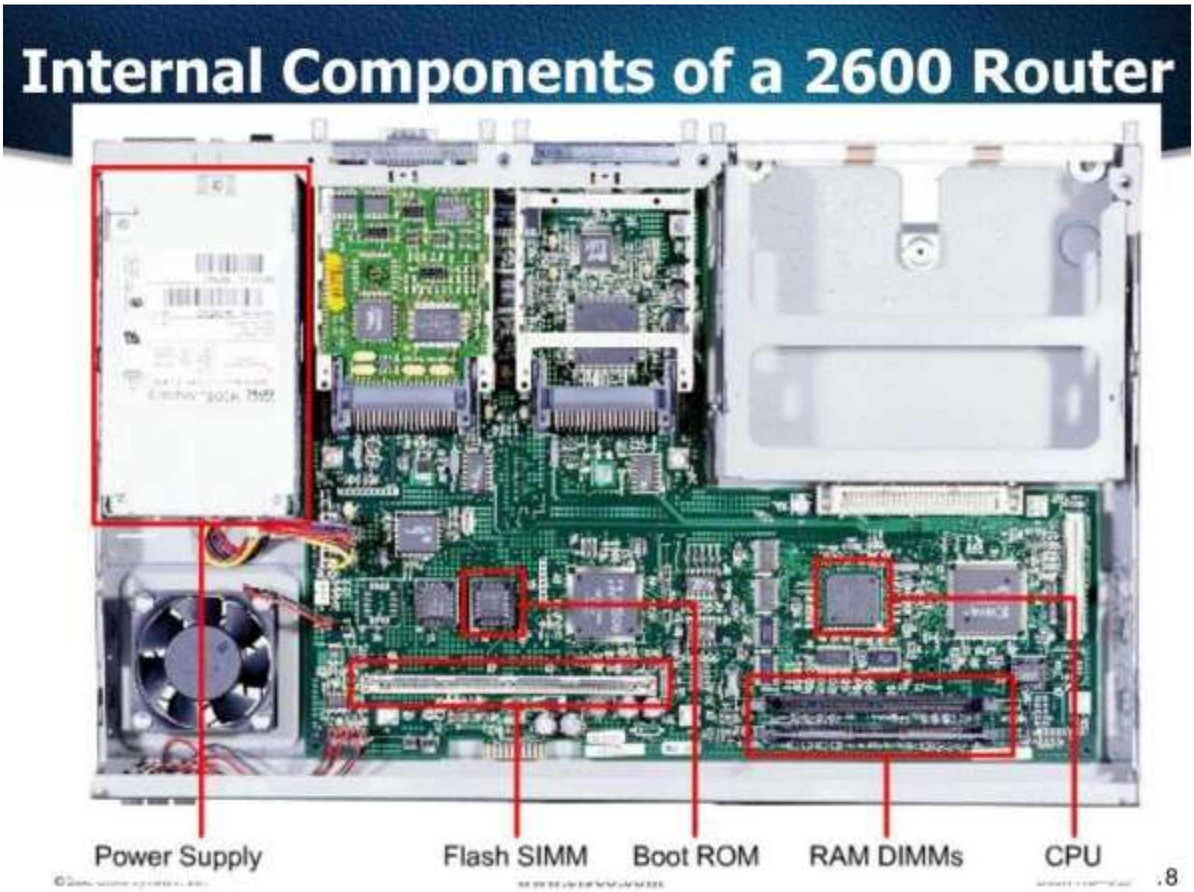
Submission Date : 26.07.2024

a. Demo session of any two networking hardware and Functionalities.
b. Implement Byte Stuffing and Bit stuffing.
c. Execute networking commands with various options and update the result screens.

a. Sample Networking Devices

- Ethernet
- RJ45 Jack
- LAN Switch
- Router

Sample



b. Byte Stuffing and Bit stuffing

Data link layer is responsible for something called Framing, which is the division of stream of bits from network layer into manageable units (called frames). Each frame consists of sender's address and a destination address.The destination address defines where the packetis to go and the sender's address helps the recipent acknowledge the receipt.

Frames could be of fixed size or variable size. In fixed-size framing, there is no need for defining the boundaries of the frames as the size itself can be used to define the end of the frame and the beginning of the next frame. But, in variable-size framing, we need a way to define the end of the frame and the beginning of the next frame.

To separate one frame from the next, an 8-bit (or 1-byte) flag is added at the beginning and the end of a frame. But the problem with that is, any pattern used for the flag could also be part of the information. So, there are two ways to overcome this problem:

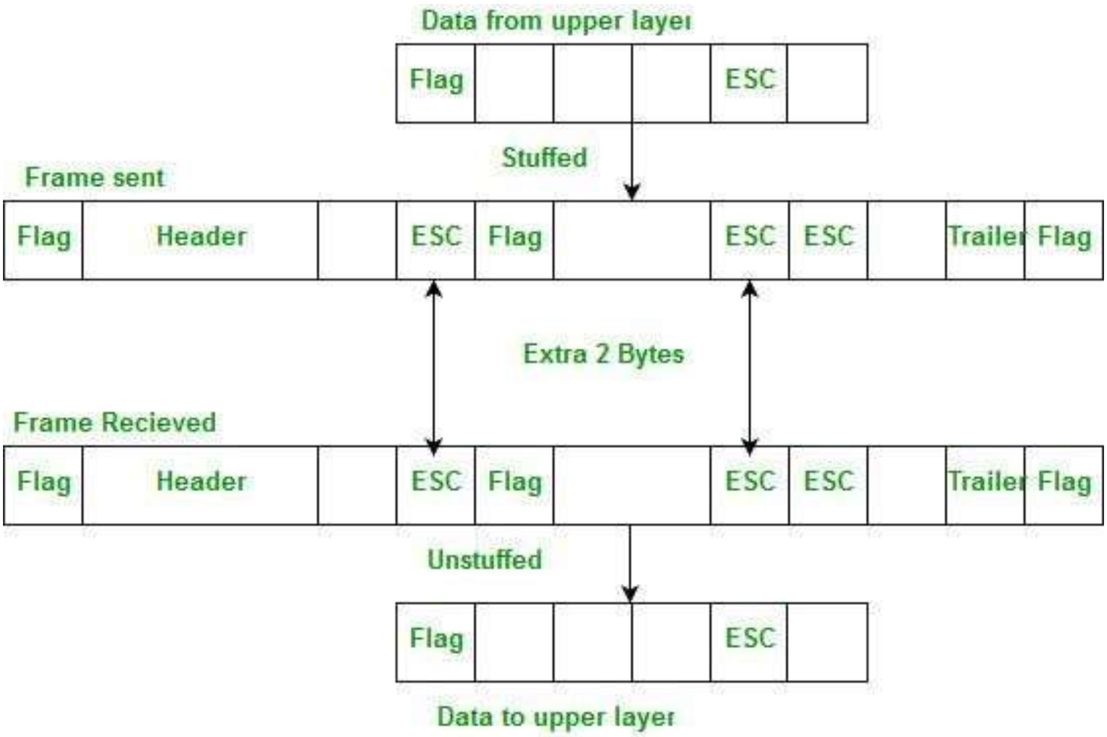
1. Using Byte stuffing (or character stuffing)
2. Using Bit stuffing

Byte stuffing –

A byte (usually escape character(ESC)), which has a predefined bit pattern is added to the data section of the frame when there is a character with the same pattern as the flag.Whenever the receiver encounters the ESC character, it removes from the data section and treats the next character as data, not a flag.

But problem arises when text contains one or more escape characters followed by a flag. To solve this problem, the escape characters that are part of the text are marked by another escape character i.e., if the escape character is part of the text,an extra one is added to show that the second one is part of the text(https://www.geeksforgeeks.org/computer-network-difference-byte-stuffing-bit-stuffing/).

Example:



Bit Stuffing Mechanism

In a data link frame, the delimiting flag sequence generally contains six or more consecutive 1s. In order to differentiate the message from the flag in case of the same sequence, a single bit is stuffed in the message. Whenever a 0 bit is followed by five consecutive 1bits in the message, an extra 0 bit is stuffed at the end of the five 1s.

When the receiver receives the message, it removes the stuffed 0s after each sequence of five 1s. The un-stuffed message is then sent to the upper layers.

Case 1

01111110 01110 01111110

01110

Case 2

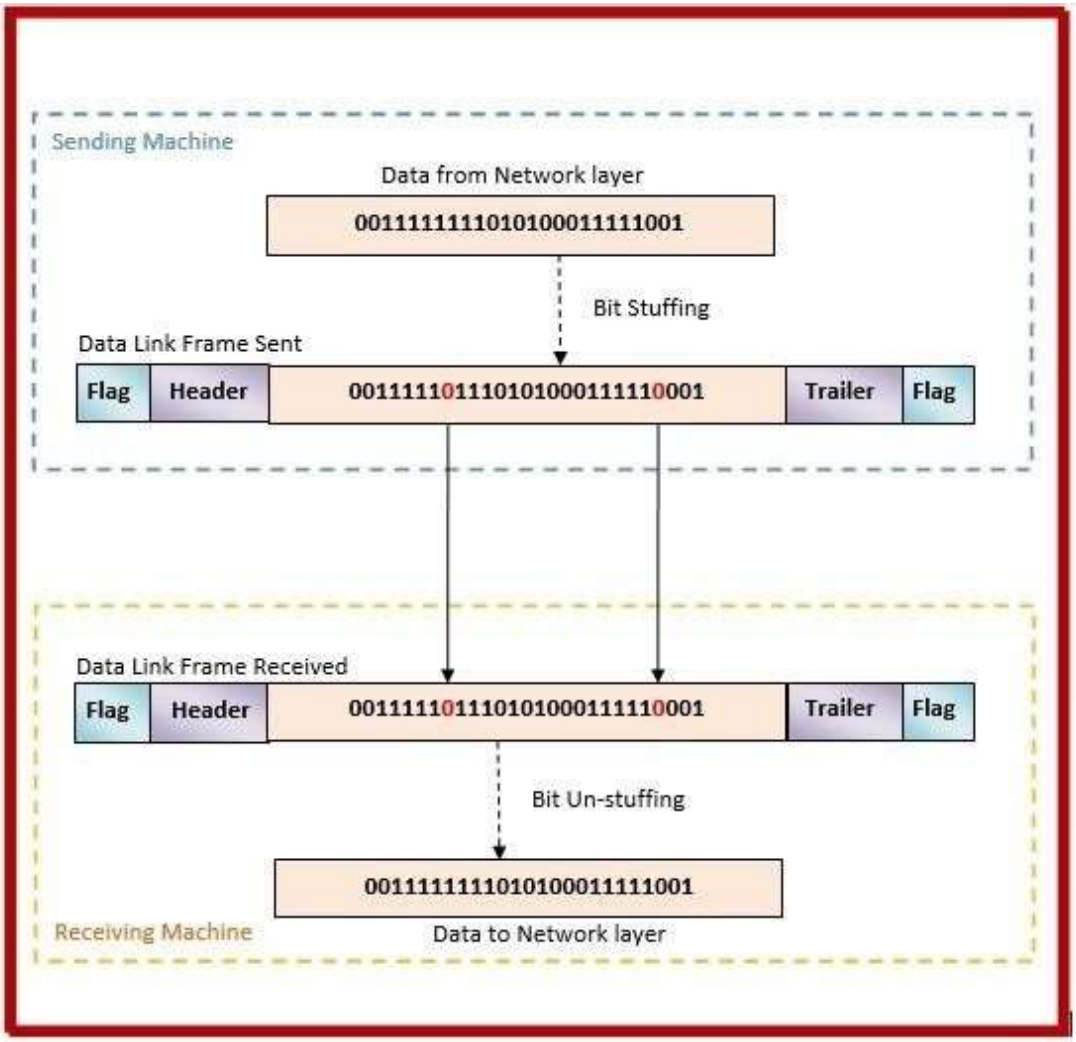
Case 2

01111110 01111110 01111110

Bit stuffing

01111110 011111010 01111110

01111110



c. Sample Networking Commands

- Ping(Packet Internet Groper)
- traceroute(Linux)
- Tracert(Windows)
- Nsloopup
- host
- Arp
- netstat
- ftp and telnet
- getmac